Smart Upgrade: Churn Prediction & Retention Strategy

Data-driven business proposal for Company A — Telecom Operator

Author: Estefany Daniela Martínez Niño
GCI World 2026 Spring | June 2026

NotebookLM

Agenda

Predictive Pipeline



The Predictive Lens: From Market Context to Measurable Value.
Data-to-Value Funnel.

NotebookLM

Colombia: A Saturated, Competitive Mobile Market

68M+
active mobile lines



Claro, Movistar
Tigo, WOM

Acquiring new customer costs 3–5× more than retaining one



ARPU: COP $45,000–65,000/month (~USD $11–16)

Postpaid churn benchmark:
18–22% annually

18M+ number portability requests by 2023 (CRC)

Cite: CRC TIC Report 2023 | MINTIC Q4 2023 | GSMA Latin America 2024

NotebookLM

Company A's Churn Rate Is 2.5× the Industry Benchmark

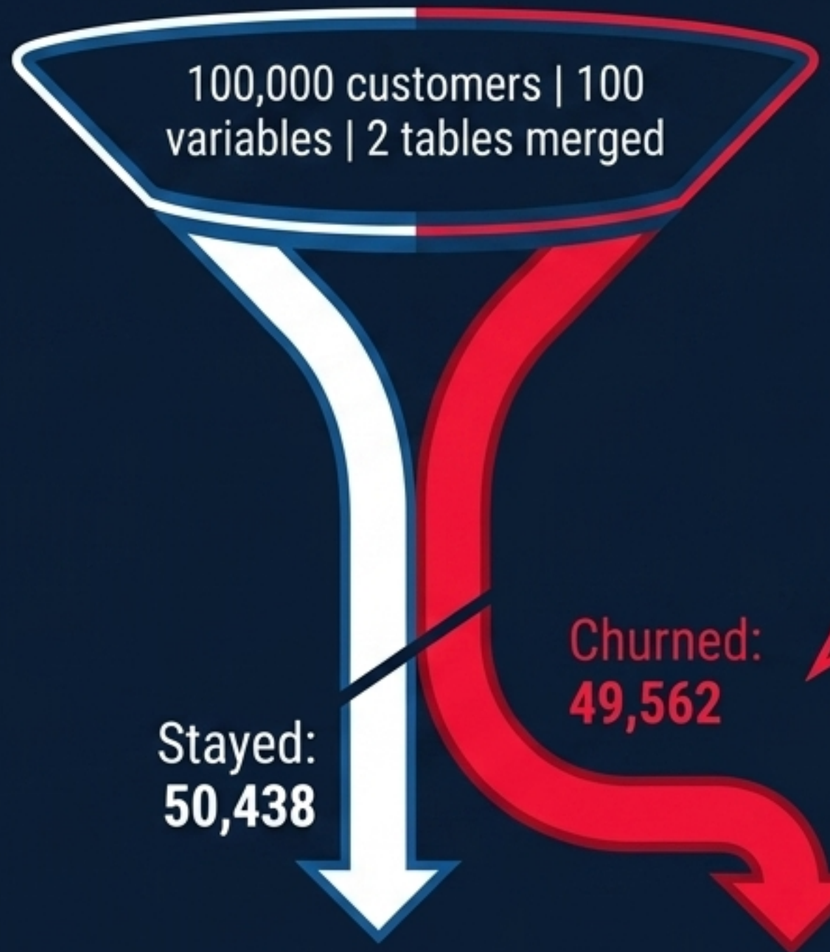
49.6%

Company A churn

2.5× gap

18–22%

Colombia benchmark

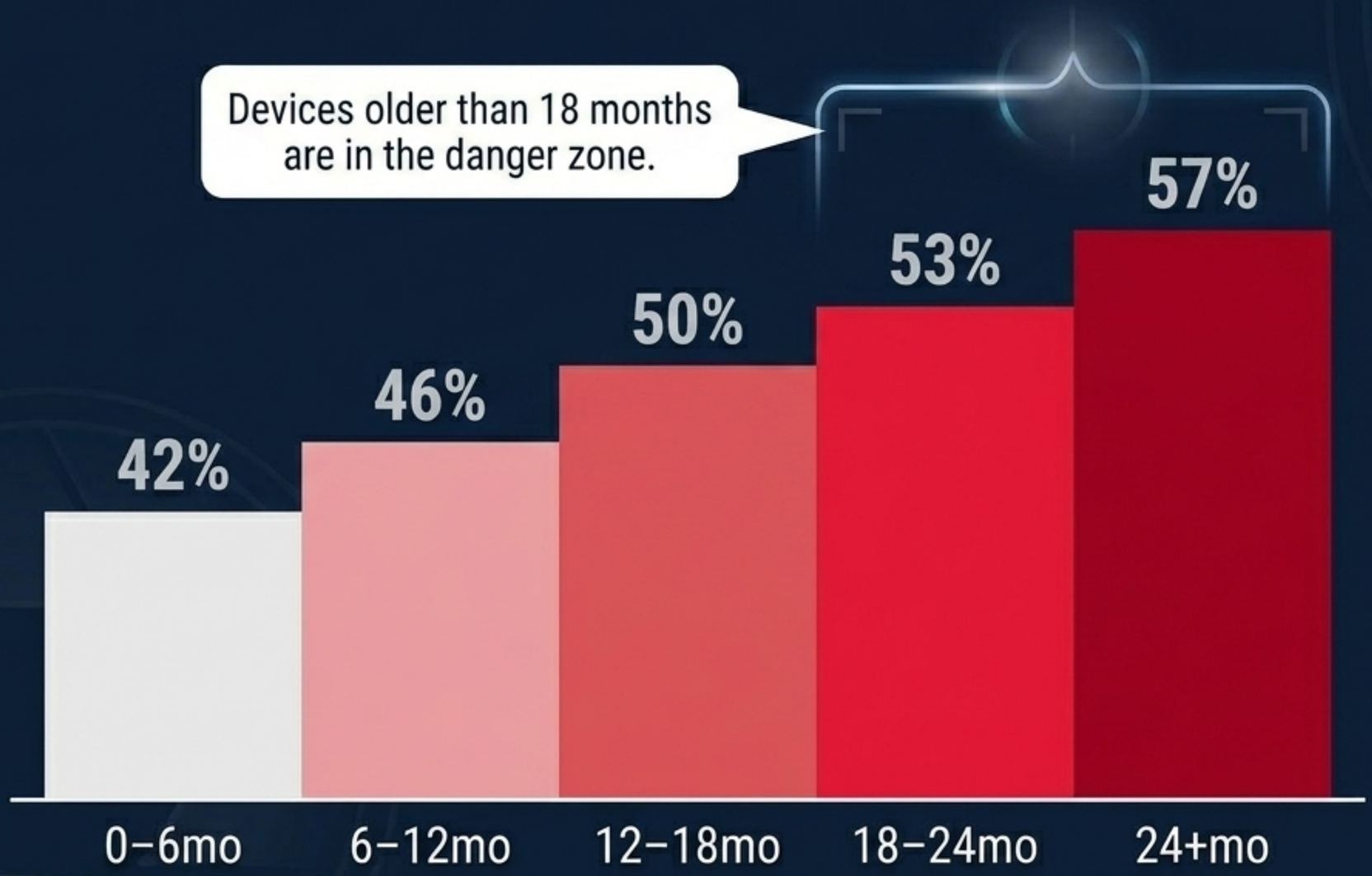


Avg monthly revenue/customer: **$58.72**

Annual revenue at risk: **$34,950,134**

NotebookLM

Older Devices = Higher Churn



Stayed avg:
363 days

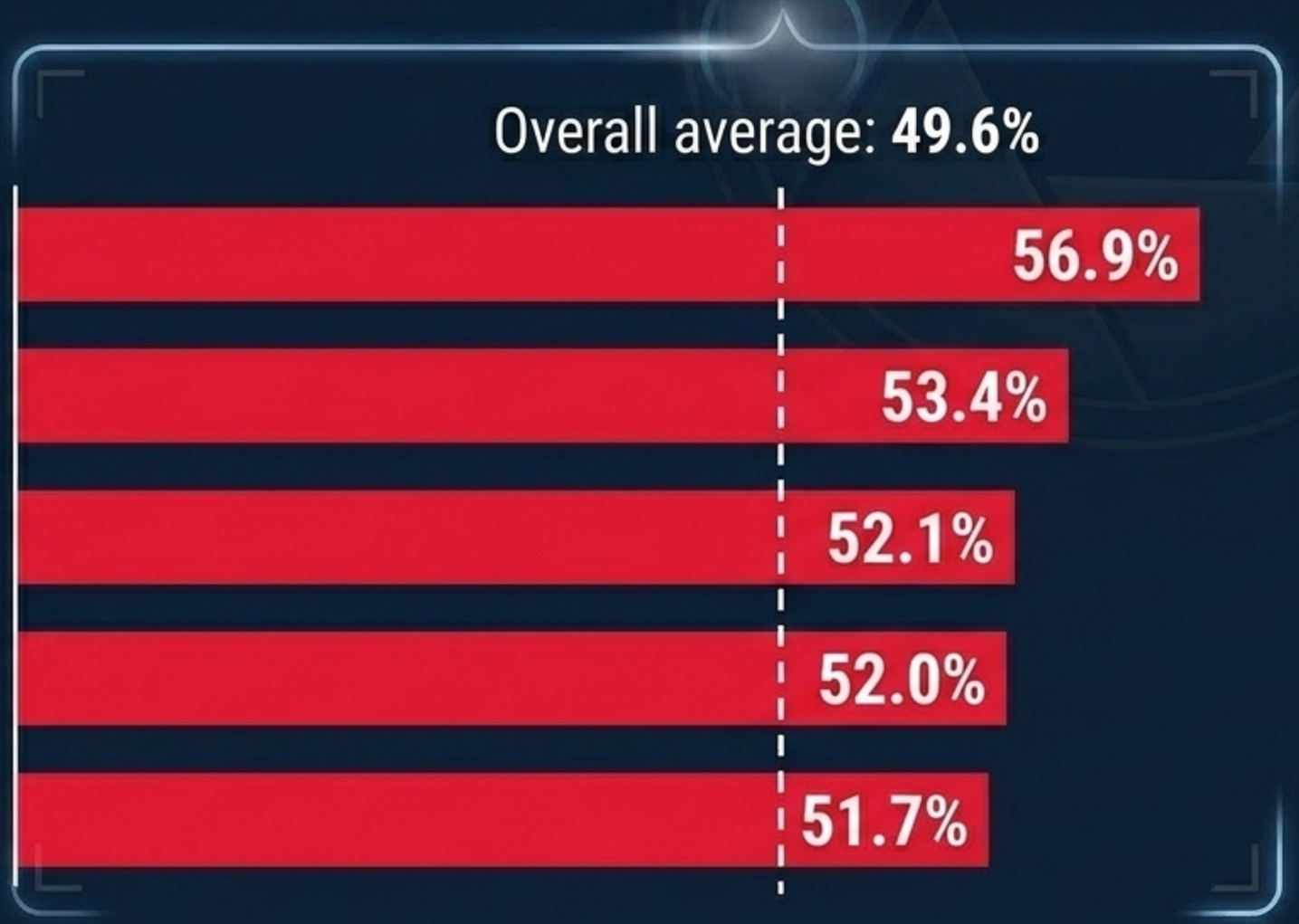
Gap: **58 days**

Churned avg:
421 days

NotebookLM

Some Regions Have Critical Churn Levels

Top 5 churn areas



NotebookLM

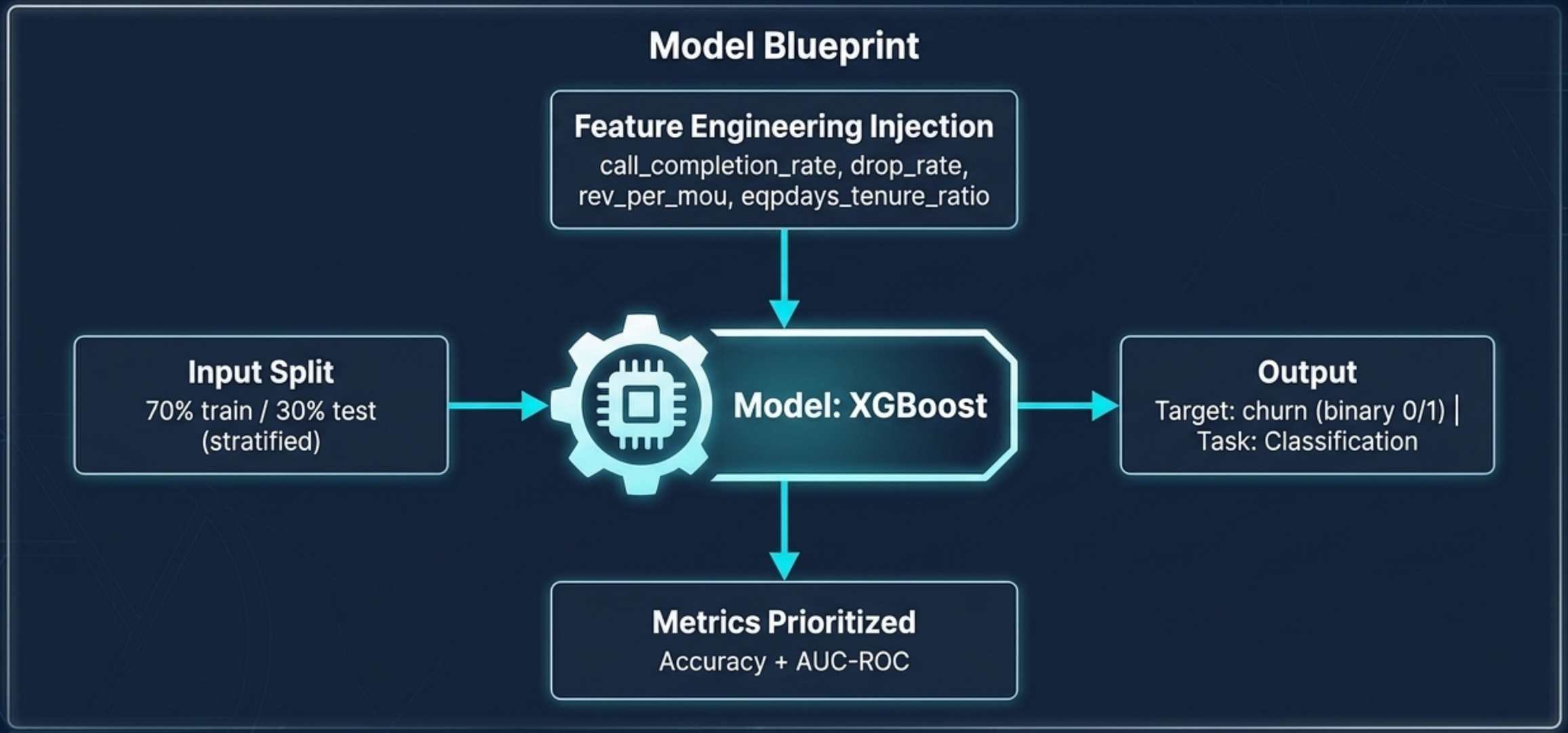
Equipment Age Is the Real Differentiator — Not Revenue

	Stayed		Churned
monthly revenue	$59.22		$58.21
custcare calls	1.98		1.60
dropped calls	6.09		5.82
eqpdays	363 days		421 days

Revenue barely differs. Device age drives churn.

NotebookLM

Which customers will churn in the next 31–60 days?



NotebookLM

XGBoost Outperforms Random Baseline by 14 Points

Performance

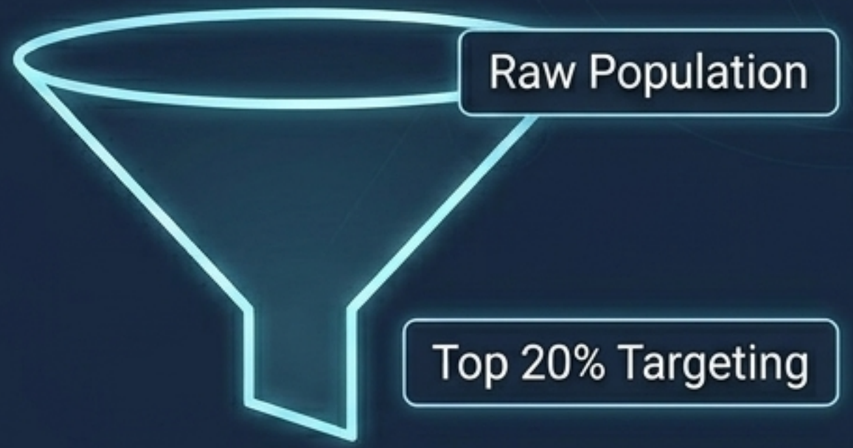
Accuracy: 63.95%

AUC-ROC: 0.6940

Baseline: ~50%

Confusion matrix (30,000 test samples):
TN: 9,540 | FP: 5,591 | FN: 5,287 | TP: 9,582

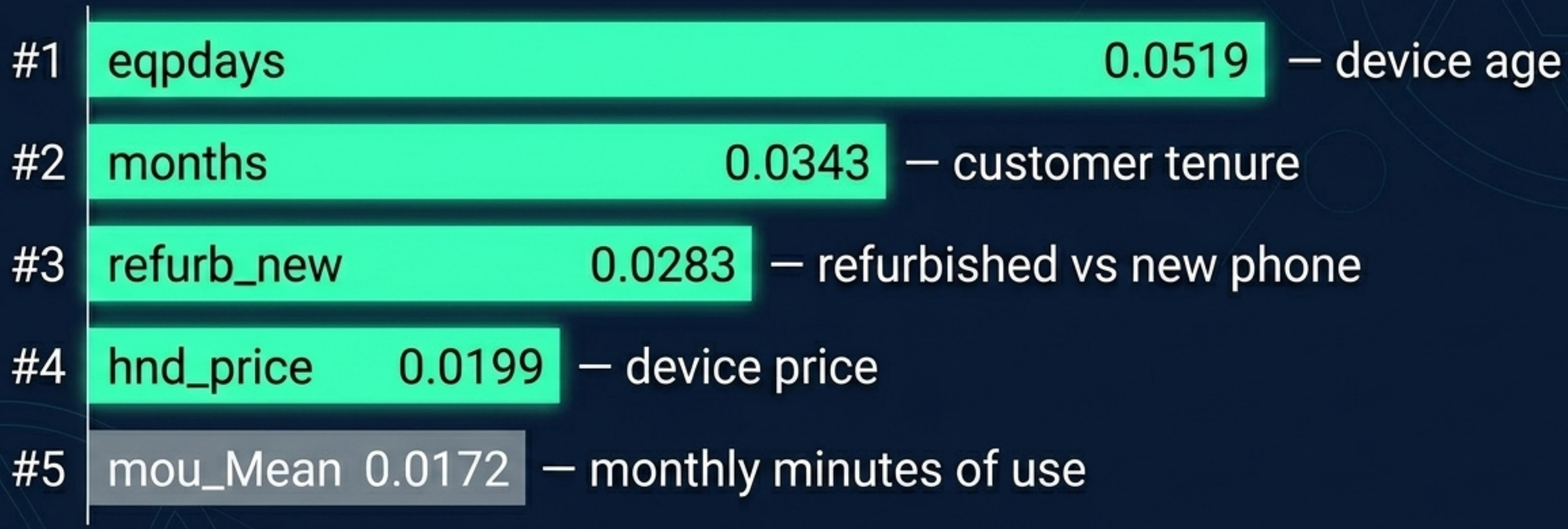
The Targeting Funnel



At top 20% targeting → precision rises to 72.97%
(3 out of 4 flagged customers actually churn)

NotebookLM

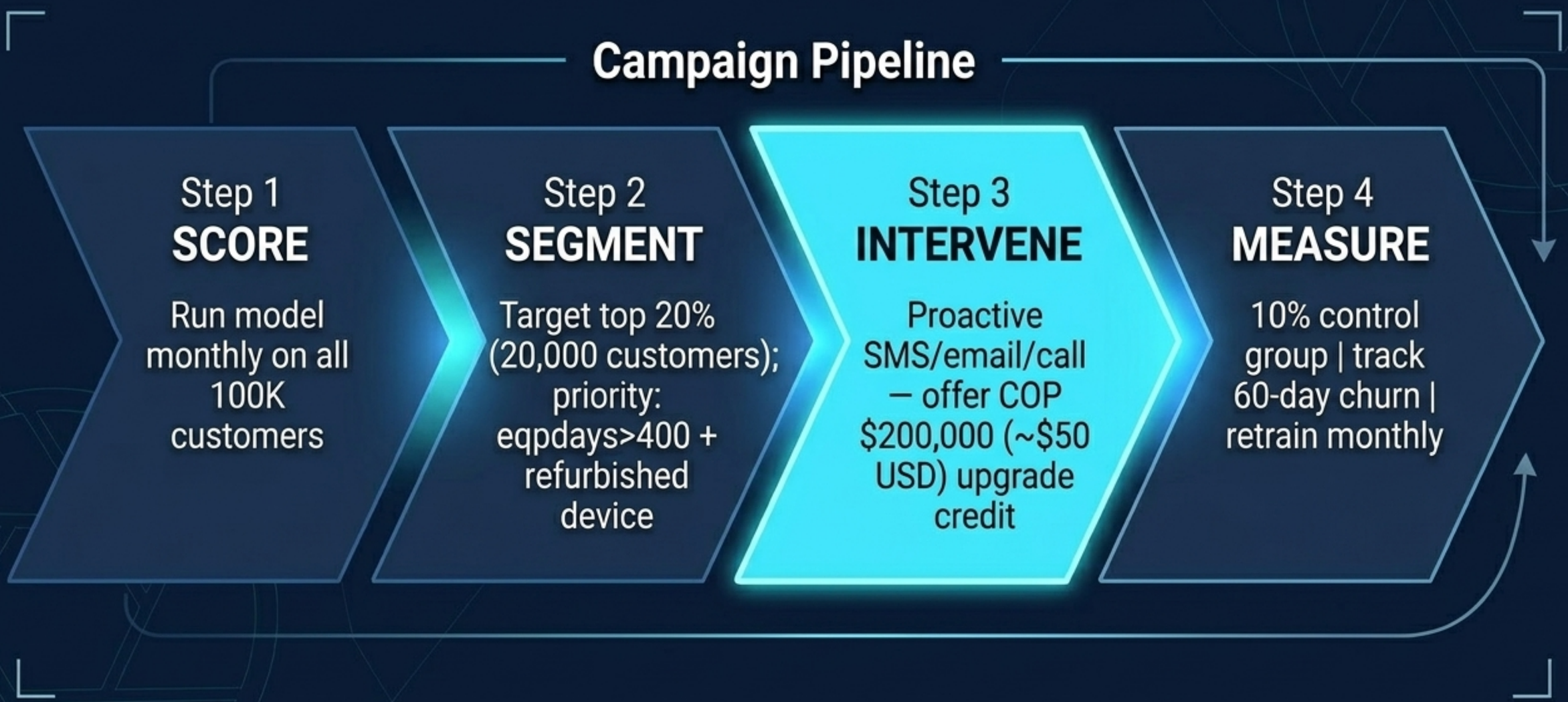
Top 4 Preeii Predictors Are All Device Signals



This confirms a **device-upgrade intervention** is the right lever.

NotebookLM

Smart Upgrade: Act Before the Customer Decides to Leave



NotebookLM

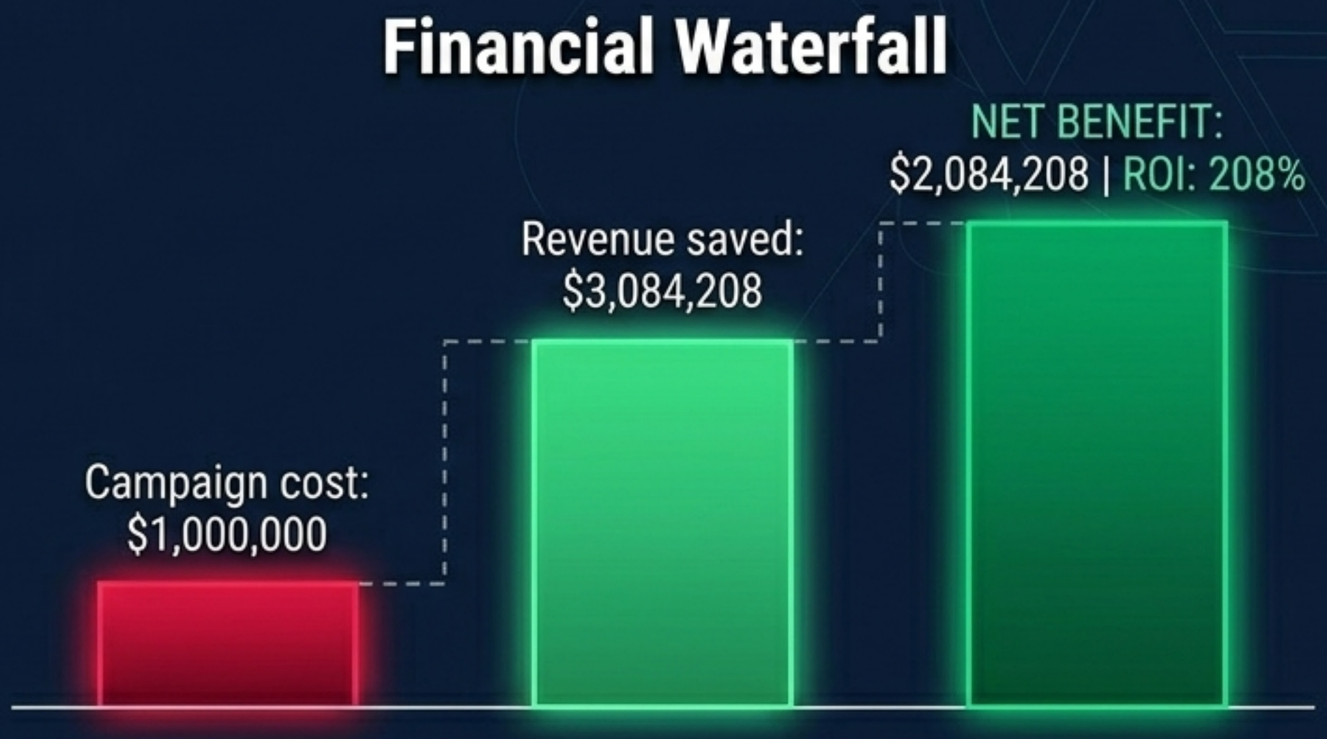
Every $1 Invested Returns $3.08 — 208% ROI

Customers targeted: 20,000

Churners in group: 14,593

Success rate assumed: 30%

Customers saved: 4,377

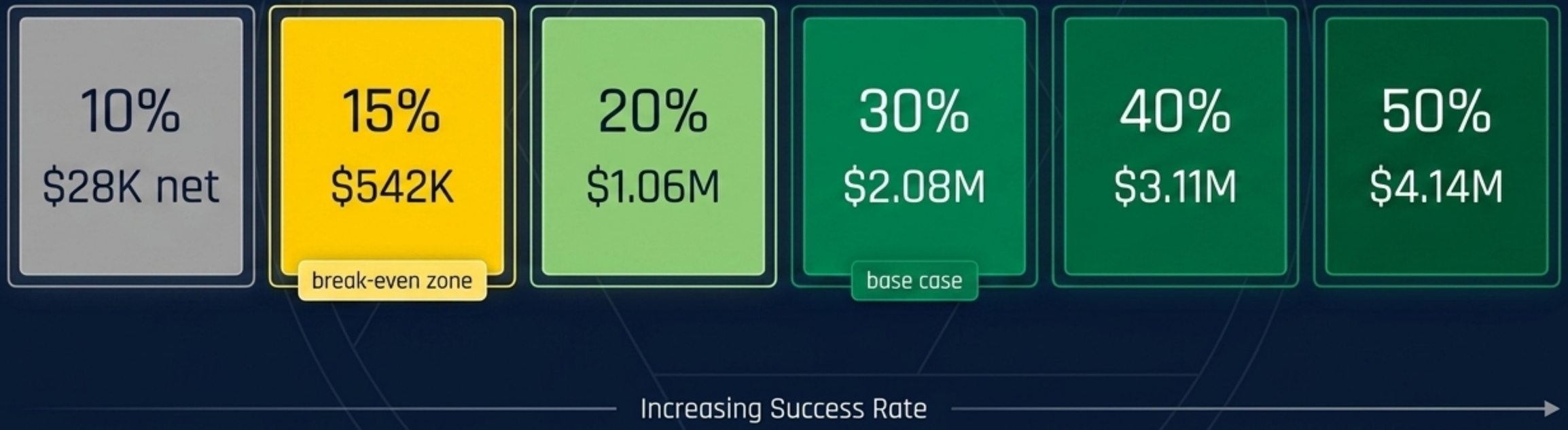


Break-even at just 14.5% success rate.

NotebookLM

PREMIUM CORPORATE INTELLIGENCE SLIDE

Profitable Even in Pessimistic Scenarios



NotebookLM

Transparency: What the Model Does Not Capture

1. Accuracy 63.95% — use probability scores not binary labels

2. 30% success rate is assumed — validate with A/B test month 1

3. Cannibalization — ~27% of targeted customers wouldn't have churned

4. Temporal bias — retrain quarterly with fresh data

5. Missing data (30–50%) in demographic columns — imputed with median

NotebookLM

References & Data Sources

[1] CRC. TIC Industry Report 2023. crcom.gov.co

[2] MINTIC. TIC Bulletin Q4 2023. mintic.gov.co

[3] GSMA. Mobile Economy Latin America 2024. gsma.com

[4] Chen & Guestrin (2016). XGBoost. ACM SIGKDD. doi:10.1145/2939672.2939785

[5] Pedregosa et al. (2011). Scikit-learn. JMLR 12, 2825–2830.

[6] GCI World 2026. Dataset Overview — Company A. [attached]

[7] GCI World 2026. Final Assignment README. [attached]

[8] Claude (Anthropic). claude.ai — [paste chat URL]

NotebookLM